

ML101

Homework 03: A ‘gentle’ Introduction to Classification and Model Selection.

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1 Introduction

This assignment introduces fundamental concepts in supervised classification and linear model selection. The primary objectives are:

- **Non-Parametric Classification:** Build intuitive foundations for the K -Nearest Neighbors (K -NN) algorithm through selected introductory literature.
- **Applied Classification:** Implement foundational classification algorithms using R/Python, covering Sections 4.7.1–4.7.5 of *An Introduction to Statistical Learning* to support upcoming laboratory work.
- **Model Selection and Regularization:** Prepare for upcoming coursework via a flipped-classroom study of Chapter 6 video lectures, focusing on subset selection, stepwise methods, and validation-based test error estimation.

2 Reading Material

To develop an intuitive understanding of the K -Nearest Neighbors (K -NN) algorithm, select and review **one** of the following introductory articles:

- “*K-Nearest Neighbor (KNN) Explained*” by Diego Lopez Yse (2023). Available at [Pinecone.com](https://pinecone.com).
- “*What is the k-nearest neighbors (KNN) algorithm?*” by Eda Kavlakoglu (2022). Available at [IBM.com](https://ibm.com).
- “*K-Nearest Neighbor (KNN) Algorithm*” by GeeksforGeeks Community (2026). Available at [GeeksforGeeks.org](https://geeksforgeeks.org).

This reading is intended for conceptual preparation; no deliverable is required for this section.

3 Lab: Classification Methods

To gain hands-on experience with classification methods, work through the Chapter 4 laboratory exercises from *An Introduction to Statistical Learning Using R* (Sections 4.7.1–4.7.5, pp. 171–181). Given the fact that all of you are pythonistas, you can cover the similar sections in the Python book.

While no formal deliverable is required for this section, working through these exercises is essential preparation for the upcoming laboratory assignment.

Table 1: Required Video Lectures for Chapter 6: Linear Model Selection and Regularization

Section	Title	Duration	Links			
6.1	Introduction and Best Subset Selection	13:45	YouTube	Video	Subtitles	Transcript
6.2	Stepwise Selection	12:27	YouTube	Video	Subtitles	Transcript
6.3	Backward Stepwise Selection	5:27	YouTube	Video	Subtitles	Transcript
6.4	Estimating test error	14:07	YouTube	Video	Subtitles	Transcript
6.5	Validation and cross validation	8:44	YouTube	Video	Subtitles	Transcript

4 Flipped Classroom

To prepare for next week’s lectures, review the assigned video recordings for Chapter 6 (“Linear Model Selection and Regularization”). These lectures are provided by Stanford Online and are hosted on their YouTube channel and the edX platform¹.

Refer to Table 1 for the required playlist. You must take handwritten notes for each lecture video, which will constitute the primary deliverable for this assignment.

5 Expected Deliverable

You will prepare a simple report sharing your handwritten lecture notes of the videos presented in section 4. Please submit your report in **PDF format** through the corresponding BrightSpace™ assignment for next week.

Happy Learning! 😊

¹Hastie, T., & Tibshirani, R. (2023). *Statistical learning* (STATSX0001) [MOOC]. edX; Stanford Online. <https://www.statlearning.com/online-courses>